

Architecture Diagrams

Four views of the five-device MCP fault-response pipeline

01 Main Pipeline

The five-device pipeline is exposed as discrete MCP tools. Every inter-stage boundary is enforced by a JSON Schema at runtime.

02 Hands Sub-Pipeline: NL → Ladder Logic

Device 4 (Hands) is itself a small compiler. A natural-language fix instruction flows through five internal stages.

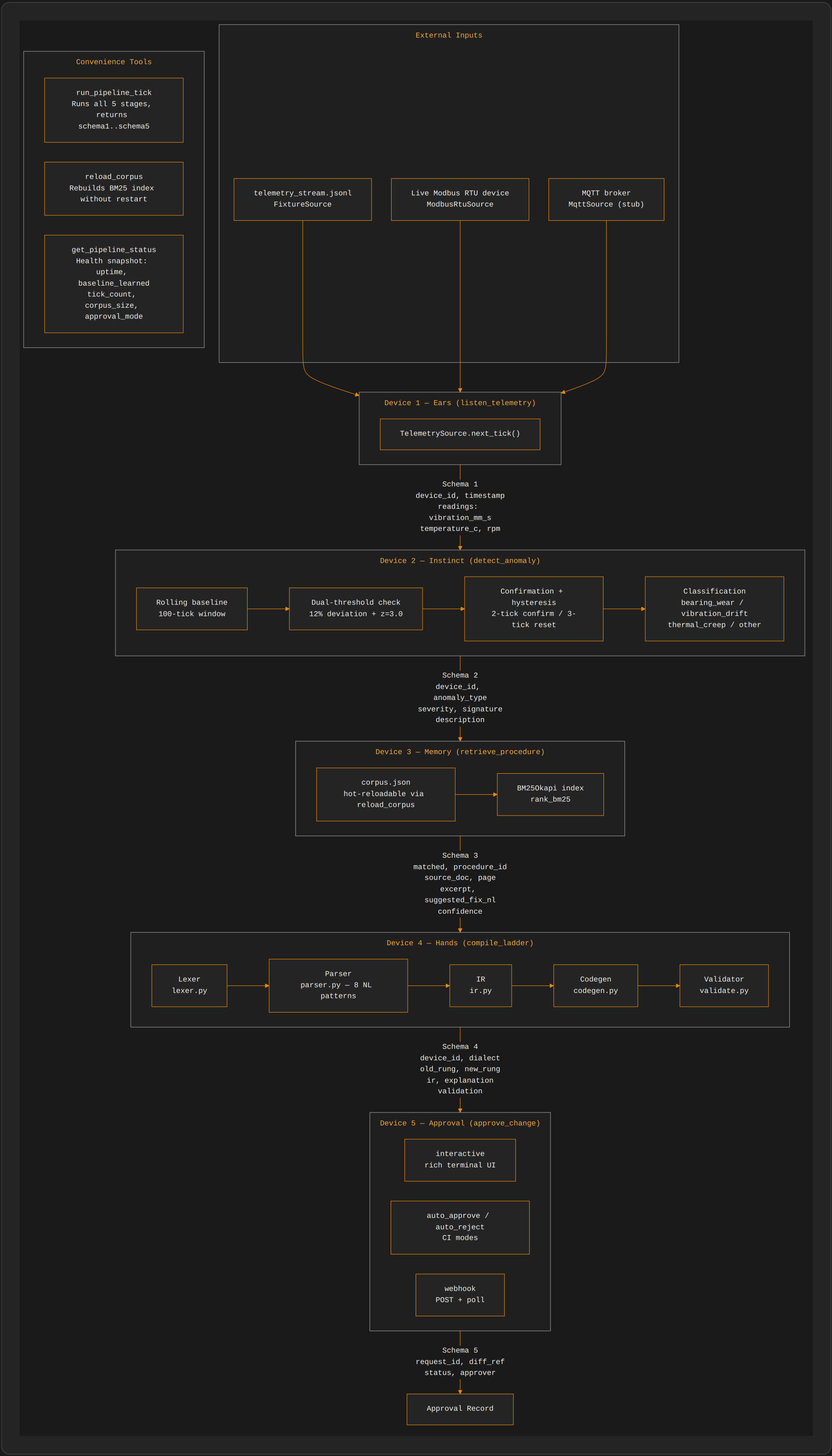
03 Pluggability Points

Two dimensions are independently swappable at startup via environment variables, with no code changes.

04 Dependency Injection Architecture

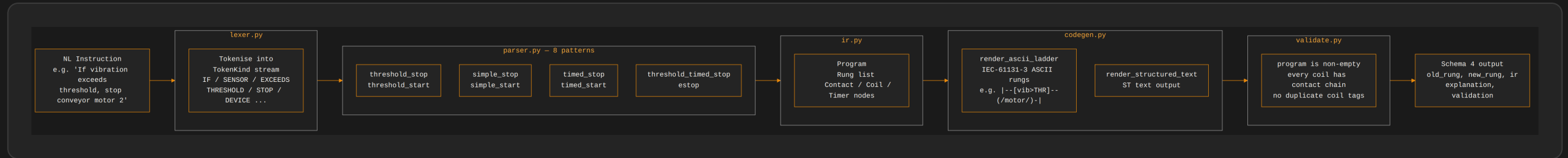
No module-level mutable singletons. All pipeline components live in a single PipelineContext dataclass, injected into every MCP tool via FastMCP's lifespan context.

Main Pipeline



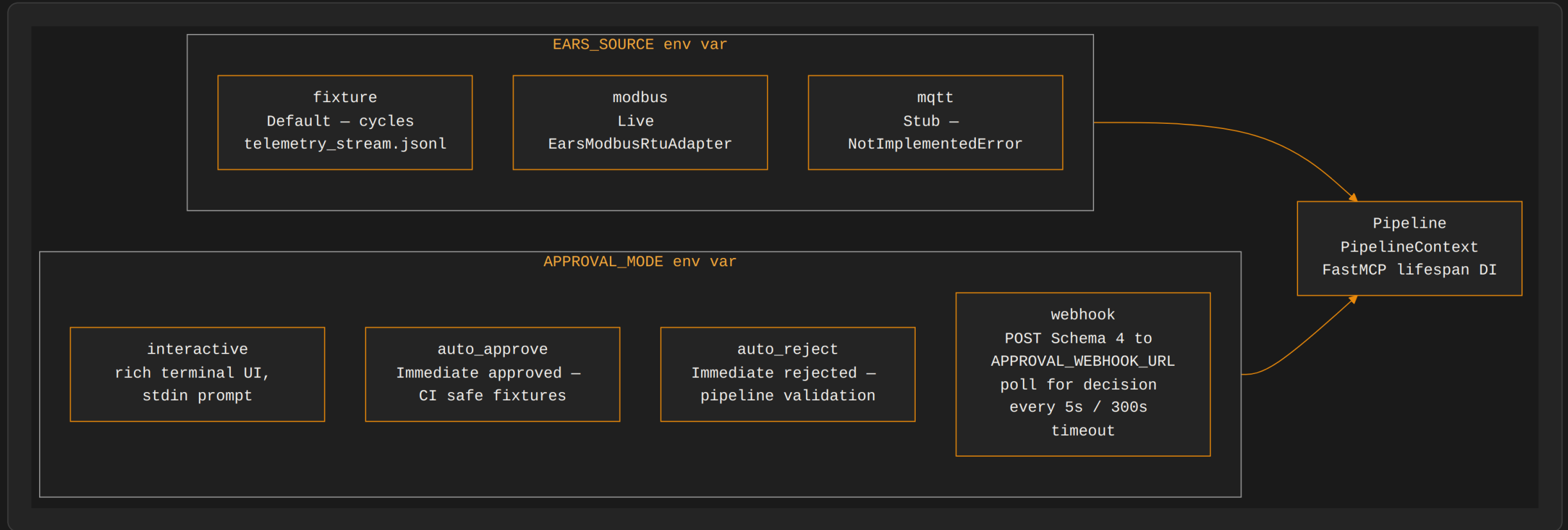
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Hands Sub-Pipeline: NL → Ladder Logic



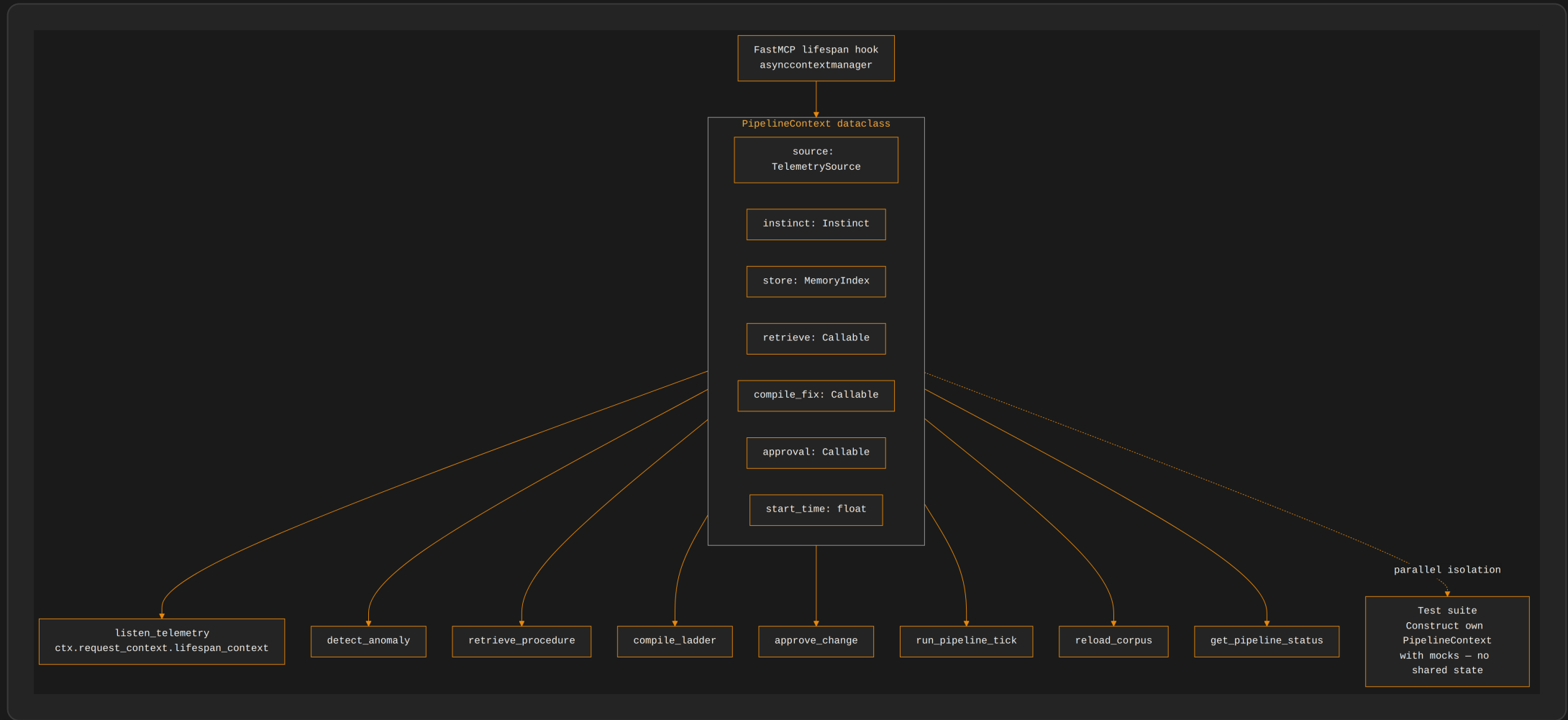
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Dependency Injection Architecture



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